

Complex Analysis

Contents

1	Complex Number System	3
1.1	Algebraic Structure	3
1.2	Conjugation and Polar Form	3
1.3	Topology and Argument	4
2	Differentiation	5
2.1	Complex Derivatives	5
2.2	Cauchy-Riemann Equations	5
2.3	Holomorphic Functions and Exponential	6
2.4	Logarithms and Branches	6
2.5	Further Holomorphicity Criteria	7
3	Series	7
3.1	Numerical Series	7
3.2	Function Series	8
3.3	Power Series	8
3.4	Consequences of Power Series	9
4	Complex Integration	10
4.1	Complex Integrals on Intervals	10
4.2	Path Integrals	10
4.3	Estimates and Uniform Convergence	13
4.4	Cauchy's Theorem	14
4.5	Cauchy's Theorem for Convex Regions	17
4.6	Cauchy's Integral Formula	18
4.7	Consequences of Cauchy's Formula	19
4.8	Cauchy Integrals and Analyticity	19
4.9	Derivative Formula	21
4.10	Operations on Power Series	22
4.11	Converse Results	23
4.12	Liouville's Theorem and Algebraic Consequences	23
5	Zeros and the Identity Theorem	24
5.1	Orders of Zeros	24
5.2	Identity Theorem	25
6	Convergence and Maximum Principles	26
6.1	Identity Theorem Applications	26
6.2	Locally Uniform Limits	26
6.3	Maximum Modulus Principle	27
6.4	Schwarz's Lemma	28
6.5	Minimum Modulus Principle	28
7	Laurent Series	29
7.1	Examples	29
7.2	Definition and Convergence	30

7.3	Annuli	32
7.4	Cauchy's Formula on an Annulus	33

1 Complex Number System

1.1 Algebraic Structure

Definition 1.1

Complex Numbers

The set of complex numbers $\mathbb{C}$ is $\mathbb{R}^2$, together with the operations

$$(x_1, y_1) + (x_2, y_2) = (x_1 + x_2, y_1 + y_2)$$

and

$$(x_1, y_1) \cdot (x_2, y_2) = (x_1x_2 - y_1y_2, x_1y_2 + x_2y_1)$$

Theorem 1.1

Field Structure

$\mathbb{C}$ is a field. With $0 = (0, 0)$, $-(x, y) = (-x, -y)$, $1 = (1, 0)$ and

$$(x, y)^{-1} = \left(\frac{x}{x^2 + y^2}, -\frac{y}{x^2 + y^2} \right) \text{ for } (x, y) \neq (0, 0)$$

Notation 1.1

If $z = (x, y)$ we call x the “real part” of z and write $x = \operatorname{Re}(z)$.

We call y the “imaginary part” of z and write $y = \operatorname{Im}(z)$.

Define $i = (0, 1)$.

Remark 1.2

There is an embedding

$$\begin{aligned} \mathbb{R} &\hookrightarrow \mathbb{C} \\ x &\mapsto (x, 0) \end{aligned}$$

so we view $\mathbb{R} \subseteq \mathbb{C}$.

Notation 1.3

Under the above embedding, any $z \in \mathbb{C}$ can be written as $z = \operatorname{Re}(z) + \operatorname{Im}(z) \cdot i$ and $i^2 = -1$

1.2 Conjugation and Polar Form

Definition 1.2

Conjugation

Define *conjugation* to be the map

$$\begin{aligned} - : \mathbb{C} &\longrightarrow \mathbb{C} \\ z = x + yi &\longmapsto x - yi = \bar{z} \end{aligned}$$

Note

$$\operatorname{Re}(z) = \frac{z + \bar{z}}{2}, \quad \operatorname{Im}(z) = \frac{z - \bar{z}}{2i}$$

Note

If $z = x + yi \in \mathbb{C}$ it has a polar representation $(x, y) \mapsto (r, \theta)$ via $z = r \cos \theta + ir \sin \theta$ where $r = \sqrt{x^2 + y^2} \geq 0$, $x = r \cos \theta$, $y = r \sin \theta$. The angle θ is defined modulo 2π when $z \neq 0$, and is arbitrary when $z = 0$.

Remark 1.4

If z_1, z_2 have polar representations

$$z_1 : (r_1, \theta_1), \quad z_2 : (r_2, \theta_2)$$

Then

$$\begin{aligned} z_1 \cdot z_2 &= (r_1 \cos \theta_1 + ir_1 \sin \theta_1) \cdot (r_2 \cos \theta_2 + ir_2 \sin \theta_2) \\ &= r_1 r_2 ((\cos \theta_1 \cos \theta_2 - \sin \theta_1 \sin \theta_2) + i(\cos \theta_1 \sin \theta_2 + \sin \theta_1 \cos \theta_2)) \\ &= r_1 r_2 (\cos(\theta_1 + \theta_2) + i \sin(\theta_1 + \theta_2)) \end{aligned}$$

1.3 Topology and Argument**Note**

If $z = x + iy$ and $r = \sqrt{x^2 + y^2}$, then $r^2 = z\bar{z}$.

Remark 1.5

$\mathbb{C}$ is a metric space and thus the triangle inequality holds.

Remark 1.6

$\mathbb{C}$ is a topological space, so topological notions make sense.

Definition 1.3**Argument**

If $z \in \mathbb{C} \setminus \{0\}$ has polar representation (r, θ) , we call θ the *argument* of z and write $\theta = \arg(z)$. We'll think about $\arg$ as a multivalued function. We write $\operatorname{Arg}(z) = \theta$ if $\theta \in (-\pi, \pi]$, the *principal value*.

Example 1.1

If $z = r(\cos \theta + i \sin \theta)$ and $z \neq 0$ then we can find an n^{th} root of z (actually n of them).

2 Differentiation

2.1 Complex Derivatives

Definition 2.1

Complex Derivative

Suppose $f : U \rightarrow \mathbb{C}$ where $U \subseteq \mathbb{C}$ is open and contains z_0 . We'll say that f has derivative $f'(z_0)$ at z_0 if

$$f'(z_0) = \lim_{z \rightarrow z_0} \frac{f(z) - f(z_0)}{z - z_0}$$

Theorem 2.1

Product Rule

If f and g are differentiable at z_0 then so is fg and

$$(f \cdot g)'(z_0) = f'(z_0)g(z_0) + f(z_0)g'(z_0)$$

2.2 Cauchy-Riemann Equations

Remark 2.1

If $f : \mathbb{C} \rightarrow \mathbb{C}$, then we may write $f(z) = u(z) + iv(z)$ where $u, v : \mathbb{C} \rightarrow \mathbb{R}$. If f is differentiable at $z_0 = x_0 + iy_0$ (say $f'(z_0) = a + ib$), then

$$\begin{aligned} R(z) &= f(z) - f(z_0) - f'(z_0)(z - z_0) \\ &= u(z) - u(z_0) - a(x - x_0) + b(y - y_0) + i(v(z) - v(z_0) - b(x - x_0) - a(y - y_0)) \\ &= R_1(z) + iR_2(z) \end{aligned}$$

Then $\lim_{z \rightarrow z_0} R_1 \frac{z}{|z - z_0|} = \lim_{z \rightarrow z_0} R_2 \frac{z}{|z - z_0|} = 0$. Thus both u and v are differentiable and $\frac{\partial u}{\partial x}(z_0) = a$, $\frac{\partial u}{\partial y}(z_0) = -b$, $\frac{\partial v}{\partial x}(z_0) = b$, $\frac{\partial v}{\partial y}(z_0) = a$.

Theorem 2.2

Cauchy-Riemann Equations

If $f : \mathbb{C} \rightarrow \mathbb{C}$, $f(z) = u(z) + iv(z)$, then f is differentiable at z_0 (as a complex function) iff both u and v are differentiable at z_0 and

$$\begin{aligned} \frac{\partial u}{\partial x}(z_0) &= \frac{\partial v}{\partial y}(z_0) \quad \text{and} \\ \frac{\partial v}{\partial x}(z_0) &= -\frac{\partial u}{\partial y}(z_0) \end{aligned}$$

These are called the Cauchy-Riemann equations.

Note

If $u : \mathbb{R}^2 \rightarrow \mathbb{R}$, then u is differentiable at (x_0, y_0) if $\frac{\partial u}{\partial x}$ and $\frac{\partial u}{\partial y}$ exist and are continuous in neighborhoods of (x_0, y_0) .

2.3 Holomorphic Functions and Exponential

Definition 2.2

Exponential function

The exponential function is defined to be the function

$$\exp(x + iy) = e^x(\cos y + i \sin y)$$

we write $e^z := \exp(z)$.

Remark 2.2

For $y \in \mathbb{R}$, we have $e^{iy} = \cos(y) + i \sin(y)$.

Definition 2.3

Holomorphic

If $f : U \rightarrow \mathbb{C}$ with $U \subseteq \mathbb{C}$ open, such that f is differentiable at each point in U , then f is *holomorphic*.

Definition 2.4

Entire

$f : \mathbb{C} \rightarrow \mathbb{C}$ is called *entire* if it is holomorphic on all of $\mathbb{C}$.

2.4 Logarithms and Branches

Definition 2.5

Logarithm

A complex number w is a *logarithm* of z if $e^w = z$.

Remark 2.3

The range of the exponential is $\mathbb{C} \setminus \{0\}$. So there is no logarithm of 0.

Definition 2.6

Branch of Logarithm

If $G \subseteq \mathbb{C} \setminus \{0\}$ is a non-empty open set, then a *branch of log* is a continuous function $\ell : G \rightarrow \mathbb{C}$ such that for all $z \in G$, $\ell(z)$ is a logarithm of z , i.e. $e^{\ell(z)} = z$

Note

A branch of log may or may not exist depending on the shape of G .

Example 2.1

$G := \mathbb{C} \setminus (-\infty, 0]$. $\ell(z) = \ln|z| + i \operatorname{Arg}(z)$. Then

$$\begin{aligned} e^{\ell(z)} &= e^{\ln|z|}(\cos(\operatorname{Arg}(z)) + i \sin(\operatorname{Arg}(z))) \\ &= |z|(\cos(\operatorname{Arg}(z)) + i \sin(\operatorname{Arg}(z))) \\ &= z \end{aligned}$$

This is called the *principal branch of log* and is denoted by $\operatorname{Log}(z)$

Note

If we define $\ell(z) = \operatorname{Log}(z) + i2\pi k$ on $\mathbb{C} \setminus (-\infty, 0]$ for fixed $k \in \mathbb{Z}$, then we get a branch of log.

Remark 2.4

It is not possible to find a branch of log on $\mathbb{C} \setminus \{0\}$.

2.5 Further Holomorphicity Criteria**Exercise 2.5**

In polar form, the Cauchy-Riemann equations become, if $f(z) = u(z) + iv(z)$

$$\begin{aligned} ru_r &= v_\theta \\ \text{and } u_\theta &= -rv_r \end{aligned}$$

Theorem 2.3**Derivative of Logarithm**

If $\ell : G \rightarrow \mathbb{C}$ is a branch of log, then ℓ is holomorphic and $\ell'(z) = \frac{1}{z}$.

Theorem 2.4**Polynomial Holomorphicity Criterion**

Suppose f is a polynomial in variables x and y (with complex coefficients), then f is holomorphic iff f can be written as polynomial in $z = x + iy$.

3 Series**3.1 Numerical Series****Definition 3.1****Series**

A *series* is a formal sum $\sum_{n=0}^{\infty} c_n$ where the terms $c_n \in \mathbb{C}$. We say a series *converges* if the sequence of partial sums converge otherwise we say it *diverges*.

Lemma 3.1**Term Test**

If $\sum_{n=0}^{\infty} c_n$ converges, then $c_n \rightarrow 0$.

Definition 3.2**Absolute Convergence**

A series $\sum_{n=0}^{\infty} c_n$ converges absolutely if $\sum_{n=0}^{\infty} |c_n| < \infty$

Lemma 3.2**Absolute Convergence Test**

If $\sum_{n=0}^{\infty} c_n$ converges absolutely, then $\sum_{n=0}^{\infty} c_n$ converges and $|\sum_{n=0}^{\infty} c_n| \leq \sum_{n=0}^{\infty} |c_n|$

Example 3.1**Geometric Series**

$\sum_{n=0}^{\infty} c^n$ converges iff $|c| < 1$. If $|c| < 1$ then $\sum_{n=0}^{\infty} c^n = \frac{1}{1-c}$

3.2 Function Series**Definition 3.3****Convergence of Functions**

Suppose $(f_n)_{n=1}^{\infty}$ is a sequence of $\mathbb{C}$ -valued functions on $G \subseteq \mathbb{C}$.

- $(f_n)_{n=1}^{\infty}$ converges pointwise to f on $S \subseteq G$ if $\lim_{n \rightarrow \infty} f_n(z) = f(z)$ for every $z \in S$.
- $(f_n)_{n=1}^{\infty}$ converges uniformly to f on S if $\sup_{z \in S} |f_n(z) - f(z)| \rightarrow 0$
- $(f_n)_{n=1}^{\infty}$ converges locally uniformly to f on G if for each $z \in G$, there is some open ball $B \subseteq G$ with centre z such that $(f_n)_{n=1}^{\infty}$ converges uniformly to f on B .

Definition 3.4**Series of Functions**

If $(f_n)_{n=0}^{\infty}$ is a sequence of functions, $\sum_{n=0}^{\infty} f_n$ is the associated series of functions. The series converges to f if the partial sums do.

3.3 Power Series**Example 3.2**

$z_0 \in \mathbb{C}$ is fixed. $(f_n)_{n=0}^{\infty}$, where $f_n(z) = a_n(z - z_0)^n$ we get the series $\sum_{n=0}^{\infty} a_n(z - z_0)^n$. The power series centred at z_0 .

Proposition 3.3**Region of Convergence**

The region of convergence for a power series $\sum_{n=0}^{\infty} a_n(z - z_0)^n$ is of the form either

1. $\{z_0\}$
2. $\mathbb{C}$
3. The points in some disk centred at z_0 and possibly points on the boundary.

Example 3.3

$\sum_{n=0}^{\infty} z^n$ has radius of convergence 1. This represents the function $\frac{1}{1-z}$ on $\mathbb{D}$. This converges locally uniformly.

Theorem 3.4**Comparison Test for Power Series**

Suppose $\sum_{n=0}^{\infty} a_n(z - z_0)^n$ and $\sum_{n=0}^{\infty} b_n(z - z_0)^n$ are power series with radius of convergence R_1, R_2 respectively. If $|b_n| \leq M|a_n|$ for some $M > 0$, then $R_1 \leq R_2$

Proof: This follows from the previous proposition because if $|z - z_0| < R$, where R is the radius of convergence, then $\sum a_n(z - z_0)^n$ converges absolutely. $\square$

Theorem 3.5**Cauchy-Hadamard Theorem**

The radius of convergence for a power series $\sum a_n(z - z_0)^n$ is

$$\frac{1}{\limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}}}$$

Theorem 3.6**Differentiation of Power Series**

Suppose $\sum_{n=0}^{\infty} a_n(z - z_0)^n$ has radius of convergence $R > 0$. Let f be the function it represents (on that disk). Then the radius of convergence for $\sum_{n=1}^{\infty} na_n(z - z_0)^{n-1}$ is also R and if g is the function it represents, then f is differentiable and $f'(z) = g(z)$ for $z \in G = \{z \in \mathbb{C} \mid |z - z_0| < R\}$

Proof: Note

$$\begin{aligned} & \frac{1}{\limsup_{n \rightarrow \infty} |na_n|^{\frac{1}{n}}} \\ &= \frac{1}{\lim_{n \rightarrow \infty} n^{\frac{1}{n}}} \cdot \frac{1}{\limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}}} \\ &= \frac{1}{\limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}}} \\ &= R \end{aligned}$$

 $\square$ **Example 3.4**

$f(z) = \sum_{n=0}^{\infty} \frac{1}{n!} z^n$ then f is entire, $f'(z) = f(z)$ and $f(0) = 1$.

3.4 Consequences of Power Series**Corollary 3.7****Smoothness of Power Series**

If f is represented by a power series on its disk of convergence, then derivatives of all orders exist on that disk.

4 Complex Integration

4.1 Complex Integrals on Intervals

Definition 4.1

Complex Integral

Suppose $\varphi : [a, b] \rightarrow \mathbb{C}$ is piecewise continuous (continuous except at finitely many points where one-sided limits exist). Then $\operatorname{Re} \varphi$ and $\operatorname{Im} \varphi$ are also piecewise continuous. Set

$$\int_a^b \varphi := \int_a^b \operatorname{Re} \varphi + i \int_a^b \operatorname{Im} \varphi$$

Note this is complex linear in φ

Theorem 4.1

Fundamental Theorem of Calculus

If φ is continuous and piecewise $\mathcal{C}^1$ on $[a, b]$, then

$$\int_a^b \varphi'(t) dt = \varphi(b) - \varphi(a)$$

Theorem 4.2

Integral Triangle Inequality

$$\left| \int_a^b \varphi(t) dt \right| \leq \int_a^b |\varphi(t)| dt$$

4.2 Path Integrals

Definition 4.2

Path

A path is a function $[a, b] \rightarrow \mathbb{C}$.

Definition 4.3

Piecewise $\mathcal{C}^1$ Path

A path $\gamma : [a, b] \rightarrow \mathbb{C}$ is piecewise $\mathcal{C}^1$ if γ is continuous and there is a partition of $[a, b]$ such that γ is $\mathcal{C}^1$ on each subinterval, with one-sided limits for γ' at the breakpoints.

Definition 4.4

Path Integral

Suppose $\gamma : [a, b] \rightarrow \mathbb{C}$ is piecewise- $\mathcal{C}^1$ and $f : G \rightarrow \mathbb{C}$ is continuous on a set G containing the image of γ . Then define

$$\int_{\gamma} f(z) dz := \int_a^b f(\gamma(t)) \gamma'(t) dt$$

Example 4.1

$f(z) = ie^x$ where $z = x + iy$, $\gamma : [a, b] \rightarrow \mathbb{C}$, $\gamma(t) = t$ then

$$\int_{\gamma} f(z) dz = \int_a^b ie^t dt = i(e^b - e^a)$$

Theorem 4.3**Fundamental Theorem for Path Integrals**

If $\gamma : [a, b] \rightarrow \mathbb{C}$ is piecewise- $\mathcal{C}^1$ and if f is holomorphic on an open set that contains the range of γ then

$$\int_{\gamma} f'(z) dz = f(\gamma(b)) - f(\gamma(a))$$

Proof:

$$\int_{\gamma} f'(z) dz = \int_a^b f'(\gamma(t))\gamma'(t) dt = \int_a^b (f \circ \gamma)'(t) dt = (f \circ \gamma)(b) - (f \circ \gamma)(a)$$

□

Proposition 4.4**Reparametrization Invariance**

If $\gamma : [a, b] \rightarrow \mathbb{C}$ is piecewise- $\mathcal{C}^1$ and $\beta : [a_1, b_1] \rightarrow [a, b]$ is increasing and piecewise- $\mathcal{C}^1$ with $\beta(a_1) = a$ and $\beta(b_1) = b$. Set $\gamma_1 = \gamma \circ \beta$. Suppose f is continuous on the image of γ . Then

$$\int_{\gamma_1} f(z) dz = \int_{\gamma} f(z) dz.$$

Proof: By definition of the path integral and the chain rule,

$$\begin{aligned} \int_{\gamma_1} f(z) dz &= \int_{a_1}^{b_1} f(\gamma_1(s))\gamma_1'(s) ds \\ &= \int_{a_1}^{b_1} f(\gamma(\beta(s)))\gamma'(\beta(s))\beta'(s) ds \end{aligned}$$

By the substitution theorem with $t = \beta(s)$,

$$\begin{aligned} \int_{a_1}^{b_1} f(\gamma(\beta(s)))\gamma'(\beta(s))\beta'(s) ds &= \int_a^b f(\gamma(t))\gamma'(t) dt \\ &= \int_{\gamma} f(z) dz \end{aligned}$$

□

Proposition 4.5**Properties of Path Integrals**

1. If $a < c < b$ and $\gamma : [a, b] \rightarrow \mathbb{C}$ is piecewise- $\mathcal{C}^1$, let $\gamma_1 = \gamma|_{[a, c]}$ and $\gamma_2 = \gamma|_{[c, b]}$. Then

$$\begin{aligned} \int_{\gamma} f(z) \, dz &= \int_a^b f(\gamma(t)) \gamma'(t) \, dt \\ &= \int_a^c f(\gamma(t)) \gamma'(t) \, dt + \int_c^b f(\gamma(t)) \gamma'(t) \, dt \\ &= \int_{\gamma_1} f(z) \, dz + \int_{\gamma_2} f(z) \, dz \end{aligned}$$

2. If $\gamma : [a, b] \rightarrow \mathbb{C}$ is piecewise- $\mathcal{C}^1$, define the reverse path $-\gamma : [a, b] \rightarrow \mathbb{C}$ by $(-\gamma)(t) = \gamma(a + b - t)$. Then

$$\begin{aligned} \int_{-\gamma} f(z) \, dz &= \int_a^b f(\gamma(a + b - t)) \cdot (-\gamma'(a + b - t)) \, dt \\ &= - \int_a^b f(\gamma(a + b - t)) \gamma'(a + b - t) \, dt \\ &= - \int_a^b f(\gamma(s)) \gamma'(s) \, ds \\ &= - \int_{\gamma} f(z) \, dz \end{aligned}$$

Example 4.2

$z_0 \in \mathbb{C}$, $R > 0$, $n \in \mathbb{Z}$, $\gamma : [0, 2\pi] \rightarrow \mathbb{C}$, $\gamma(t) = z_0 + Re^{it}$. If $n \neq -1$, then

$$\int_{\gamma} (z - z_0)^n dz = 0$$

If $n = -1$, then

$$\int_{\gamma} (z - z_0)^n dz = 2\pi i$$

Indeed, if $n \neq -1$, then $F(z) = \frac{1}{n+1}(z - z_0)^{n+1}$ is an antiderivative of $(z - z_0)^n$, so

$$\begin{aligned} \int_{\gamma} (z - z_0)^n dz &= F(\gamma(2\pi)) - F(\gamma(0)) \\ &= \frac{1}{n+1}(\gamma(2\pi) - z_0)^{n+1} - \frac{1}{n+1}(\gamma(0) - z_0)^{n+1} \\ &= 0. \end{aligned}$$

If $n = -1$, then

$$\begin{aligned} \int_{\gamma} \frac{1}{z - z_0} dz &= \int_0^{2\pi} \frac{1}{\gamma(t) - z_0} \gamma'(t) dt \\ &= \int_0^{2\pi} \frac{1}{Re^{it}} iRe^{it} dt \\ &= \int_0^{2\pi} i dt \\ &= 2\pi i. \end{aligned}$$

4.3 Estimates and Uniform Convergence**Definition 4.5****Length of a Path**

If $\gamma : [a, b] \rightarrow \mathbb{C}$ is piecewise- $\mathcal{C}^1$ its length is

$$L(\gamma) := \int_a^b |\gamma'(t)| dt$$

Theorem 4.6**ML Inequality**

If f is continuous on the image of γ , then

$$\left| \int_{\gamma} f(z) dz \right| \leq L(\gamma) M_f$$

where M_f is the maximum value of $|(f \circ \gamma)(t)|$ for $t \in [a, b]$.

Proof:

$$\begin{aligned}
 \left| \int_{\gamma} f(z) \, dz \right| &= \left| \int_a^b f(\gamma(t)) \gamma'(t) \, dt \right| \\
 &\leq \int_a^b |f(\gamma(t)) \gamma'(t)| \, dt \\
 &= \int_a^b |f(\gamma(t))| |\gamma'(t)| \, dt \\
 &\leq \int_a^b M_f |\gamma'(t)| \, dt \\
 &= M_f L(\gamma)
 \end{aligned}$$

□

Theorem 4.7

Uniform Convergence for Path Integrals

Suppose $\gamma : [a, b] \rightarrow \mathbb{C}$ is piecewise- $\mathcal{C}^1$ and $(f_n)_{n=1}^{\infty}$ are continuous on the image of γ and (f_n) converges uniformly to f on the range of γ . Then

$$\int_{\gamma} f = \lim_{n \rightarrow \infty} \int_{\gamma} f_n$$

Proof: Since $f_n \rightarrow f$ uniformly and each f_n is continuous on the image of γ , f is continuous on the image of γ . By the ML inequality,

$$\begin{aligned}
 \left| \int_{\gamma} f_n - \int_{\gamma} f \right| &= \left| \int_{\gamma} (f_n - f) \right| \\
 &\leq L(\gamma) \sup_{t \in [a, b]} |f_n(\gamma(t)) - f(\gamma(t))| \\
 &= L(\gamma) \sup_{z \in \gamma([a, b])} |f_n(z) - f(z)|.
 \end{aligned}$$

Since $f_n \rightarrow f$ uniformly on the range of γ , the last expression tends to 0. Hence $\int_{\gamma} f_n \rightarrow \int_{\gamma} f$. □

4.4 Cauchy's Theorem

Example 4.3

Fourier Transform of the Gaussian

Let $a, b > 0$ and let $R_{a,b}$ be the positively oriented boundary of the rectangle with vertices $-a$, a , $a + ib$, and $-a + ib$. Consider

$$\int_{R_{a,b}} e^{-z^2} \, dz.$$

Since

$$e^{-z^2} = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n}}{n!},$$

define

$$g(z) = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n+1}}{(2n+1)n!}.$$

The power series for g has infinite radius of convergence, and termwise differentiation gives $g'(z) = e^{-z^2}$. Hence

$$\int_{R_{a,b}} e^{-z^2} dz = \int_{R_{a,b}} g'(z) dz = 0.$$

Writing the four sides of $R_{a,b}$ separately gives

$$0 = \int_{-a}^a e^{-x^2} dx + \int_0^b e^{-(a+iy)^2} i dy - \int_{-a}^a e^{-(x+ib)^2} dx - \int_0^b e^{-(-a+iy)^2} i dy.$$

Thus

$$\int_{-a}^a e^{-(x+ib)^2} dx = \int_{-a}^a e^{-x^2} dx + i \int_0^b e^{-(a+iy)^2} dy - i \int_0^b e^{-(-a+iy)^2} dy.$$

Since

$$e^{-(x+ib)^2} = e^{-x^2+b^2-2ibx},$$

we get

$$\int_{-a}^a e^{-x^2} e^{-2ibx} dx = e^{-b^2} \int_{-a}^a e^{-x^2} dx + i e^{-b^2} \int_0^b e^{-(a+iy)^2} dy - i e^{-b^2} \int_0^b e^{-(-a+iy)^2} dy.$$

For $0 \leq y \leq b$,

$$|e^{-(a+iy)^2}| = e^{-a^2+y^2} \leq e^{-a^2+b^2}$$

and the same bound holds for $e^{-(-a+iy)^2}$. Hence the two vertical integrals tend to 0 as $a \rightarrow \infty$.

Therefore

$$\int_{-\infty}^{\infty} e^{-x^2} e^{-2ibx} dx = e^{-b^2} \int_{-\infty}^{\infty} e^{-x^2} dx.$$

Since $\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi}$,

$$\int_{-\infty}^{\infty} e^{-x^2} e^{-2ibx} dx = \sqrt{\pi} e^{-b^2}.$$

Theorem 4.8**Cauchy's Theorem for a Triangle (Goursat's Lemma)**

If T is a triangle in $\mathbb{C}$ and if f is holomorphic in an open region that contains T and its interior, then

$$\int_T f(z) \, dz = 0.$$

Proof: Let

$$I = \int_T f(z) \, dz$$

and assume that T is positively oriented. Divide T into four congruent subtriangles by joining the midpoints of the sides. The boundary integrals over the four subtriangles add to I , because the integrals over the shared sides cancel. Hence one subtriangle, call it T_1 , satisfies

$$\left| \int_{T_1} f(z) \, dz \right| \geq \frac{|I|}{4}.$$

Repeating this construction gives nested triangles

$$T \supseteq T_1 \supseteq T_2 \supseteq \dots$$

such that

$$\left| \int_{T_n} f(z) \, dz \right| \geq \frac{|I|}{4^n}.$$

If P is the perimeter of T and D is the diameter of T , then the perimeter and diameter of T_n are $\frac{P}{2^n}$ and $\frac{D}{2^n}$ respectively.

Since the triangles are nested compact sets and their diameters tend to 0, there is a unique point

$$z_0 \in \bigcap_{n=1}^{\infty} T_n.$$

Since f is holomorphic at z_0 ,

$$f(z) = f(z_0) + f'(z_0)(z - z_0) + \eta(z)(z - z_0),$$

where $\eta(z) \rightarrow 0$ as $z \rightarrow z_0$. The first two terms have primitives, so their integrals around T_n are 0. Thus

$$\int_{T_n} f(z) \, dz = \int_{T_n} \eta(z)(z - z_0) \, dz.$$

Let $\varepsilon > 0$. Since $\eta(z) \rightarrow 0$, for all sufficiently large n we have $|\eta(z)| < \varepsilon$ for all $z \in T_n$. Hence, by the ML inequality,

$$\left| \int_{T_n} f(z) dz \right| \leq \varepsilon \cdot \left(\frac{D}{2^n} \right) \cdot \left(\frac{P}{2^n} \right) = \frac{\varepsilon PD}{4^n}.$$

Combining this with the lower bound gives, for all sufficiently large n ,

$$\frac{|I|}{4^n} \leq \left| \int_{T_n} f(z) dz \right| \leq \frac{\varepsilon PD}{4^n}.$$

Multiplying by 4^n , we get $|I| \leq \varepsilon PD$. Since $\varepsilon > 0$ was arbitrary, $I = 0$. □

4.5 Cauchy's Theorem for Convex Regions

Theorem 4.9

Cauchy's Theorem for Convex Regions

Suppose f is holomorphic in G and G is convex, then for any piecewise- $\mathcal{C}^1$ closed curve γ in G we have

$$\int_{\gamma} f(z) dz = 0$$

Proof: Fix $z_0 \in G$. For $z \in G$, define

$$F(z) = \int_{[z_0, z]} f(w) dw.$$

This is well-defined since G is convex. We claim that $F' = f$.

Let $z \in G$. Since G is open, for h sufficiently small we have $z + h \in G$. By convexity, the triangle with vertices $z_0, z, z + h$ is contained in G , so by Cauchy's theorem for triangles,

$$\int_{[z_0, z]} f(w) dw + \int_{[z, z+h]} f(w) dw - \int_{[z_0, z+h]} f(w) dw = 0.$$

Hence

$$F(z+h) - F(z) = \int_{[z, z+h]} f(w) dw.$$

Parametrizing the line segment $[z, z+h]$ by $w = z + th$, $0 \leq t \leq 1$, gives

$$\frac{F(z+h) - F(z)}{h} = \int_0^1 f(z+th) dt.$$

Therefore

$$\frac{F(z+h) - F(z)}{h} - f(z) = \int_0^1 (f(z+th) - f(z)) dt.$$

Since f is continuous at z , the right side tends to 0 as $h \rightarrow 0$. Thus $F'(z) = f(z)$.

Now if $\gamma : [a, b] \rightarrow G$ is a piecewise- $\mathcal{C}^1$ closed curve, then by [Theorem 4.3](#),

$$\int_{\gamma} f(z) dz = F(\gamma(b)) - F(\gamma(a)) = 0,$$

since $\gamma(a) = \gamma(b)$. □

4.6 Cauchy's Integral Formula

Theorem 4.10

Cauchy's Formula for the Circle

Suppose C is a circle oriented counter-clockwise and f is holomorphic in a region that contains C and its interior, then if z is in the interior of C

$$f(z) = \frac{1}{2\pi i} \int_C \frac{f(\zeta)}{\zeta - z} d\zeta$$

Proof: Let

$$g(\zeta) = \begin{cases} \frac{f(\zeta) - f(z)}{\zeta - z} & \text{if } \zeta \neq z \\ f'(z) & \text{if } \zeta = z. \end{cases}$$

Then g is holomorphic in the interior of C , so by Cauchy's theorem for convex regions,

$$\int_C g(\zeta) d\zeta = 0.$$

Hence

$$\int_C \frac{f(\zeta)}{\zeta - z} d\zeta = f(z) \int_C \frac{1}{\zeta - z} d\zeta.$$

It remains to compute the last integral. Write $C(t) = z_0 + Re^{it}$, $0 \leq t \leq 2\pi$, and set $a = \frac{z - z_0}{R}$. Since z is in the interior of C , $|a| < 1$. Then

$$\begin{aligned} \int_C \frac{1}{\zeta - z} d\zeta &= \int_0^{2\pi} \frac{iRe^{it}}{z_0 + Re^{it} - z} dt \\ &= \int_0^{2\pi} \frac{i}{1 - ae^{-it}} dt \\ &= \int_0^{2\pi} i \sum_{n=0}^{\infty} a^n e^{-int} dt \\ &= 2\pi i. \end{aligned}$$

The third equality uses the geometric series expansion, valid uniformly since $|a| < 1$. The last equality follows by integrating the uniformly convergent series term-by-term: all terms with $n \geq 1$ integrate to 0, and the $n = 0$ term gives $2\pi i$. Therefore

$$\int_C \frac{f(\zeta)}{\zeta - z} d\zeta = 2\pi i f(z),$$

which gives the result. □

Note

The integral is 0 for z outside the circle

4.7 Consequences of Cauchy's Formula

Corollary 4.11

Mean Value Property

Suppose f is holomorphic on a region containing $\{z \in \mathbb{C} \mid |z - z_0| < R\}$. If $0 < r < R$, then

$$f(z_0) = \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + re^{it}) dt.$$

Proof: Let C_r be the circle of radius r centered at z_0 , oriented counter-clockwise. By Cauchy's formula,

$$f(z_0) = \frac{1}{2\pi i} \int_{C_r} \frac{f(\zeta)}{\zeta - z_0} d\zeta.$$

Parametrize C_r by $\gamma(t) = z_0 + re^{it}$ for $0 \leq t \leq 2\pi$. Then $\gamma'(t) = ire^{it}$, so

$$\begin{aligned} f(z_0) &= \frac{1}{2\pi i} \int_0^{2\pi} \frac{f(z_0 + re^{it})}{re^{it}} ire^{it} dt \\ &= \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + re^{it}) dt. \end{aligned}$$

□

Note

This is called the *mean value property* because $f(z_0)$ is equal to the average of the values of f on the circle centered at z_0 .

4.8 Cauchy Integrals and Analyticity

Definition 4.6

If φ is a continuous function on the range of some piecewise- $\mathcal{C}^1$ path γ . The *Cauchy Integral* over γ is the function

$$z \mapsto \int_{\gamma} \frac{\varphi(\zeta)}{\zeta - z} d\zeta$$

This is defined for z not in the range of γ

Proposition 4.12**Power Series of a Cauchy Integral**

Let γ be a piecewise- $\mathcal{C}^1$ path and suppose φ is continuous on the range of γ . Define

$$f(z) = \int_{\gamma} \frac{\varphi(\zeta)}{\zeta - z} d\zeta$$

for z not in the range of γ . If z_0 is not in the range of γ and

$$R = \text{dist}(z_0, \text{range}(\gamma)),$$

then f has a power series representation on $\{z \in \mathbb{C} \mid |z - z_0| < R\}$.

Proof: Let z satisfy $|z - z_0| < R$. Then for ζ in the range of γ ,

$$\left| \frac{z - z_0}{\zeta - z_0} \right| < 1.$$

Thus

$$\begin{aligned} \frac{1}{\zeta - z} &= \frac{1}{\zeta - z_0 - (z - z_0)} \\ &= \frac{1}{\zeta - z_0} \cdot \frac{1}{1 - \frac{z - z_0}{\zeta - z_0}} \\ &= \sum_{n=0}^{\infty} \frac{(z - z_0)^n}{(\zeta - z_0)^{n+1}}. \end{aligned}$$

Therefore

$$f(z) = \int_{\gamma} \varphi(\zeta) \sum_{n=0}^{\infty} \frac{(z - z_0)^n}{(\zeta - z_0)^{n+1}} d\zeta.$$

If $0 < \rho < R$ and $|z - z_0| \leq \rho$, then

$$\left| \frac{z - z_0}{\zeta - z_0} \right| \leq \frac{\rho}{R} < 1$$

for all ζ in the range of γ . Hence the geometric series converges uniformly on $|z - z_0| \leq \rho$ and on the range of γ , so we may integrate term by term. Thus

$$f(z) = \sum_{n=0}^{\infty} \left(\int_{\gamma} \frac{\varphi(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta \right) (z - z_0)^n.$$

This is a power series centered at z_0 . □

Theorem 4.13**Holomorphic Functions are Analytic**

Suppose f is holomorphic on a region containing $\{z \in \mathbb{C} \mid |z - z_0| < R\}$. Then f has a power series representation centered at z_0 with radius of convergence at least R . More precisely, for $0 < r < R$ and $|z - z_0| < r$,

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n,$$

where

$$a_n = \frac{1}{2\pi i} \int_{C_r} \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta,$$

and C_r is the positively oriented circle of radius r centered at z_0 .

Proof: Fix $0 < r < R$. By Cauchy's formula, for $|z - z_0| < r$,

$$f(z) = \frac{1}{2\pi i} \int_{C_r} \frac{f(\zeta)}{\zeta - z} d\zeta.$$

This is a Cauchy integral with $\varphi(\zeta) = \frac{f(\zeta)}{2\pi i}$. By the power series representation for Cauchy integrals,

$$f(z) = \sum_{n=0}^{\infty} \left(\int_{C_r} \frac{f(\zeta)}{2\pi i (\zeta - z_0)^{n+1}} d\zeta \right) (z - z_0)^n.$$

Hence

$$a_n = \frac{1}{2\pi i} \int_{C_r} \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta.$$

Since $r < R$ was arbitrary, the radius of convergence is at least R . □

4.9 Derivative Formula**Corollary 4.14****Cauchy's Integral Formula for Derivatives**

Suppose f is holomorphic on a region containing $\{z \in \mathbb{C} \mid |z - z_0| < R\}$. If $0 < r < R$ and C_r is the positively oriented circle of radius r centered at z_0 , then for every $n \geq 0$,

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \int_{C_r} \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta.$$

Proof: By the previous theorem, for $|z - z_0| < r$,

$$f(z) = \sum_{n=0}^{\infty} a_n (z - z_0)^n,$$

where

$$a_n = \frac{1}{2\pi i} \int_{C_r} \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta.$$

Since power series can be differentiated term-by-term inside their radius of convergence,

$$f^{(n)}(z_0) = n!a_n.$$

Substituting the formula for a_n gives

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \int_{C_r} \frac{f(\zeta)}{(\zeta - z_0)^{n+1}} d\zeta.$$

□

Corollary 4.15

Holomorphic Derivatives

If f is holomorphic on a region G , then $f^{(n)}$ is holomorphic on G for every $n \geq 0$.

Proof: Let $z_0 \in G$. Since G is open, there is $R > 0$ such that $\{z \in \mathbb{C} \mid |z - z_0| < R\} \subseteq G$. By the theorem above, f has a power series representation on this disk. Power series can be differentiated term-by-term inside their radius of convergence, so f' is holomorphic on the disk. Repeating this argument gives that $f^{(n)}$ is holomorphic near z_0 for every $n \geq 0$. Since $z_0 \in G$ was arbitrary, $f^{(n)}$ is holomorphic on G . □

4.10 Operations on Power Series

Proposition 4.16

Cauchy Product of Power Series

Suppose $\sum_{n=0}^{\infty} a_n(z - z_0)^n$ has radius of convergence R_1 and $\sum_{n=0}^{\infty} b_n(z - z_0)^n$ has radius of convergence R_2 . Consider

$$\sum_{n=0}^{\infty} \left(\sum_{k=0}^n a_k b_{n-k} \right) (z - z_0)^n$$

Then the radius of convergence for this power series is at least $\min\{R_1, R_2\}$. If f and g are the functions defined by the first two power series, then for $|z - z_0| < \min\{R_1, R_2\}$,

$$f(z)g(z) = \sum_{n=0}^{\infty} \left(\sum_{k=0}^n a_k b_{n-k} \right) (z - z_0)^n.$$

Proof: Let $R = \min\{R_1, R_2\}$ and define $h(z) = f(z)g(z)$ for $|z - z_0| < R$. Since f and g are holomorphic on this disk, so is h . Thus h has a power series representation

$$h(z) = \sum_{n=0}^{\infty} c_n(z - z_0)^n$$

on $|z - z_0| < R$, where $c_n = h^{(n)}(z_0)/n!$. By Leibniz's rule,

$$\begin{aligned}
h^{(n)}(z_0) &= \sum_{k=0}^n \binom{n}{k} f^{(k)}(z_0) g^{(n-k)}(z_0) \\
&= \sum_{k=0}^n \frac{n!}{k!(n-k)!} \cdot k! a_k \cdot (n-k)! b_{n-k} \\
&= n! \sum_{k=0}^n a_k b_{n-k}.
\end{aligned}$$

Hence

$$c_n = \sum_{k=0}^n a_k b_{n-k}.$$

Therefore the Cauchy product represents $h = fg$ on $|z - z_0| < R$, so its radius of convergence is at least R . □

4.11 Converse Results

Proposition 4.17

Converse of Goursat's Lemma

If $f : G \rightarrow \mathbb{C}$ is continuous and $\int_T f = 0$ for every triangle T whose edges and interior are contained in G , then f is holomorphic.

Proof: It is enough to show that f is holomorphic in a neighbourhood of each point. Fix $z_0 \in G$ and choose a disk D centered at z_0 with $D \subseteq G$.

For $z \in D$, define

$$g(z) = \int_{[z_0, z]} f(\zeta) d\zeta.$$

The triangle hypothesis implies that this integral is path independent in D : if $z_1, z_2, z_3 \in D$, then the integral around the triangle with these vertices is 0.

By the same line-segment argument as in the proof of Cauchy's theorem for convex regions, $g' = f$ on D . Therefore g is holomorphic on D . Since holomorphic derivatives are holomorphic, $f = g'$ is holomorphic on D . Since z_0 was arbitrary, f is holomorphic on G . □

4.12 Liouville's Theorem and Algebraic Consequences

Theorem 4.18

Liouville's Theorem

Suppose $f : \mathbb{C} \rightarrow \mathbb{C}$ is entire and bounded. Then f is constant.

Proof: Suppose $|f(z)| \leq M$ for all $z \in \mathbb{C}$. Fix $z_0 \in \mathbb{C}$ and $r > 0$. Let C_r be the circle centered at z_0 with radius r , oriented counter-clockwise. By Cauchy's integral formula for derivatives,

$$f'(z_0) = \frac{1}{2\pi i} \int_{C_r} \frac{f(\zeta)}{(\zeta - z_0)^2} d\zeta.$$

By the ML inequality,

$$|f'(z_0)| \leq \frac{1}{2\pi} \cdot \left(\frac{M}{r^2} \right) \cdot 2\pi r = \frac{M}{r}.$$

Since this holds for every $r > 0$, letting $r \rightarrow \infty$ gives $f'(z_0) = 0$. Since z_0 was arbitrary, $f' = 0$ on $\mathbb{C}$, so f is constant. $\square$

Theorem 4.19

The Fundamental Theorem of Algebra

If $p(z) = \sum_{k=0}^n a_k z^k$ is a nonconstant polynomial with complex coefficients and $a_n \neq 0$, then there exist $z_1, \dots, z_n \in \mathbb{C}$ such that

$$p(z) = a_n \prod_{k=1}^n (z - z_k).$$

Proof: It is enough to show that p has a zero, since after finding a zero z_1 , we may factor $p(z) = (z - z_1)q(z)$ and repeat by induction on the degree.

Suppose, for contradiction, that p has no zero. Then $\frac{1}{p}$ is entire. We claim that $\frac{1}{p}$ is bounded. Since $a_n \neq 0$,

$$p(z) = z^n \left(a_n + \frac{a_{n-1}}{z} + \dots + \frac{a_0}{z^n} \right).$$

For $|z|$ sufficiently large, the term in parentheses has absolute value at least $\frac{|a_n|}{2}$, so

$$|p(z)| \geq \frac{|a_n|}{2} \cdot |z|^n.$$

Hence $\left| \frac{1}{p(z)} \right| \rightarrow 0$ as $|z| \rightarrow \infty$, so $\frac{1}{p}$ is bounded outside a large disk. On the closed disk, $\frac{1}{p}$ is continuous, hence bounded. Thus $\frac{1}{p}$ is bounded on $\mathbb{C}$.

By Liouville's theorem, $\frac{1}{p}$ is constant, so p is constant, contradicting that p is nonconstant. Therefore p has a zero, and the factorization follows by induction. $\square$

5 Zeros and the Identity Theorem

5.1 Orders of Zeros

Definition 5.1

Zero of Order n

Suppose f is holomorphic on G and $f(z_0) = 0$. We say z_0 is a *zero of order n* if

$$0 = f(z_0) = f'(z_0) = \dots = f^{(n-1)}(z_0)$$

but $f^{(n)}(z_0) \neq 0$. A zero of order 1 is called a *simple zero*.

Proposition 5.1

Infinite-Order Zero

Suppose G is open, f is holomorphic on G , and f has a zero of infinite order at $z_0 \in G$. Then $f = 0$ on the connected component of G containing z_0 .

Proof: Let H be the connected component of G containing z_0 . Define

$$A = \{z \in H \mid f \text{ is identically zero on some disk centered at } z\}.$$

Since f has a zero of infinite order at z_0 , every coefficient in the power series for f centered at z_0 is 0. Thus f is identically zero on some disk centered at z_0 , so $z_0 \in A$ and A is non-empty.

The set A is open in H by definition. We show that A is closed in H . Suppose $z_k \in A$ and $z_k \rightarrow z \in H$. For each $m \geq 0$, since f is identically zero near each z_k , we have

$$f^{(m)}(z_k) = 0.$$

Since $f^{(m)}$ is continuous, $f^{(m)}(z) = 0$. Hence all derivatives of f vanish at z , so the power series for f centered at z is identically zero on some disk centered at z . Thus $z \in A$.

Therefore A is non-empty, open, and closed in H . Since H is connected, $A = H$, so $f = 0$ on H . $\square$

Proposition 5.2

Finite-Order Zeros are Isolated

Suppose f is holomorphic on an open set G and z_0 is a zero of finite order n . Then there is a disk centered at z_0 on which z_0 is the only zero of f .

Proof: Since z_0 is a zero of order n , the power series for f centered at z_0 has the form

$$f(z) = \sum_{k=n}^{\infty} a_k (z - z_0)^k$$

with $a_n \neq 0$. Thus

$$f(z) = (z - z_0)^n g(z),$$

where

$$g(z) = \sum_{k=0}^{\infty} a_{n+k} (z - z_0)^k.$$

Then g is holomorphic near z_0 and $g(z_0) = a_n \neq 0$. By continuity, $g(z) \neq 0$ on some disk centered at z_0 . On this disk, the only zero of $f(z) = (z - z_0)^n g(z)$ is z_0 . $\square$

5.2 Identity Theorem

Corollary 5.3

Identity Theorem

Suppose f and g are holomorphic on an open set G . If f and g agree on a subset $S \subseteq G$ that has a limit point $z_0 \in G$, then $f = g$ on the connected component of G containing z_0 .

Proof: Let $h = f - g$. Then h is holomorphic on G and $h = 0$ on S . Since z_0 is a limit point of S , there is a sequence (z_n) in S with $z_n \rightarrow z_0$ and $z_n \neq z_0$. Since h is continuous, $h(z_0) = 0$.

If h had a zero of finite order at z_0 , then z_0 would be an isolated zero of h , contradicting that $z_n \rightarrow z_0$ with $z_n \neq z_0$ and $h(z_n) = 0$. Hence h has a zero of infinite order at z_0 . By the infinite-order zero proposition, $h = 0$ on the connected component of G containing z_0 . Thus $f = g$ there. $\square$

6 Convergence and Maximum Principles

6.1 Identity Theorem Applications

Example 6.1

Suppose $f : \mathbb{D} \rightarrow \mathbb{C}$ is holomorphic and $f\left(\frac{1}{n}\right) = \frac{1}{n^2}$ for all $n \in \mathbb{N}$. What is $f'\left(\frac{i}{2}\right)$?

Proof: Let $g(z) = z^2$. Then $f\left(\frac{1}{n}\right) = g\left(\frac{1}{n}\right)$ for all $n \in \mathbb{N}$, and $\frac{1}{n} \rightarrow 0 \in \mathbb{D}$. By the identity theorem, $f = g$ on $\mathbb{D}$. Hence

$$f'\left(\frac{i}{2}\right) = g'\left(\frac{i}{2}\right) = 2 \cdot \frac{i}{2} = i.$$

□

6.2 Locally Uniform Limits

Proposition 6.1

Weierstrass Theorem

Suppose $G \subseteq \mathbb{C}$ is open and $f_k : G \rightarrow \mathbb{C}$ is holomorphic for each $k \in \mathbb{N}$. If $f_k \rightarrow f$ locally uniformly on G , then f is holomorphic. Moreover, for each $n \geq 0$, $f_k^{(n)} \rightarrow f^{(n)}$ locally uniformly on G .

Proof: Fix $z_0 \in G$ and choose $R > 0$ such that the closed disk $\overline{B(z_0, R)} \subseteq G$. Let C be the positively oriented circle centered at z_0 with radius R .

If $z \in B(z_0, R)$, then by Cauchy's formula,

$$f_k(z) = \frac{1}{2\pi i} \int_C f_k \frac{\zeta}{\zeta - z} d\zeta.$$

Since $f_k \rightarrow f$ uniformly on C , the uniform convergence theorem for path integrals gives

$$f(z) = \lim_{k \rightarrow \infty} f_k(z) = \frac{1}{2\pi i} \int_C \frac{f(\zeta)}{\zeta - z} d\zeta.$$

Thus f is a Cauchy integral on $B(z_0, R)$, so f is holomorphic there. Since z_0 was arbitrary, f is holomorphic on G .

If $0 < r < R$ and $z \in B(z_0, r)$, then by Cauchy's integral formula for derivatives,

$$f_k^{(n)}(z) - f^{(n)}(z) = \frac{n!}{2\pi i} \int_C \frac{f_k(\zeta) - f(\zeta)}{(\zeta - z)^{n+1}} d\zeta.$$

Since $|\zeta - z| \geq R - r$ for $\zeta \in C$ and $z \in B(z_0, r)$, the ML inequality gives

$$\left| f_k^{(n)}(z) - f^{(n)}(z) \right| \leq n! \frac{R}{(R - r)^{n+1}} \sup_{\zeta \in C} |f_k(\zeta) - f(\zeta)|.$$

The supremum tends to 0 by local uniform convergence, so $f_k^{(n)} \rightarrow f^{(n)}$ uniformly on $B(z_0, r)$. Since z_0 was arbitrary, the convergence is locally uniform on G . □

6.3 Maximum Modulus Principle

Theorem 6.2

Maximum Modulus Principle

If $G \subseteq \mathbb{C}$ is a region and $f : G \rightarrow \mathbb{C}$ is holomorphic and nonconstant, then $|f|$ has no local maximum in G .

Proof: Suppose $|f|$ has a local maximum at $z_0 \in G$. Choose $R > 0$ such that $\{z \in \mathbb{C} \mid |z - z_0| < R\} \subseteq G$ and

$$|f(z)| \leq |f(z_0)|$$

for all $|z - z_0| < R$.

If $f(z_0) = 0$, then $|f(z)| = 0$ on this disk, so f is locally constant. By the identity theorem, f is constant on G , a contradiction. Thus $f(z_0) \neq 0$.

For $0 < r < R$, the mean value property gives

$$f(z_0) = \frac{1}{2\pi} \int_0^{2\pi} f(z_0 + re^{it}) dt.$$

Taking absolute values,

$$|f(z_0)| \leq \frac{1}{2\pi} \int_0^{2\pi} |f(z_0 + re^{it})| dt \leq |f(z_0)|.$$

Hence equality holds throughout. Since the continuous function

$$t \mapsto |f(z_0)| - |f(z_0 + re^{it})|$$

is nonnegative and has integral 0, it is identically 0. Thus $|f(z_0 + re^{it})| = |f(z_0)|$ for all t .

Equality in the triangle inequality also forces all values $f(z_0 + re^{it})$ to have the same argument as $f(z_0)$. Therefore

$$f(z_0 + re^{it}) = f(z_0)$$

for all t . Since this holds for every $0 < r < R$, $f = f(z_0)$ on the disk centered at z_0 with radius R . By the identity theorem, f is constant on G , contradicting the hypothesis. Therefore $|f|$ has no local maximum in G . □

Corollary 6.3

If $f : G \rightarrow \mathbb{C}$ is holomorphic, nonconstant and $K \subseteq G$ compact and nonempty, then $|f|$ attains its maximum on ∂K .

6.4 Schwarz's Lemma

Theorem 6.4

Schwarz's Lemma

Suppose $f : \mathbb{D} \rightarrow \mathbb{D}$ is holomorphic and $f(0) = 0$. Then $|f(z)| \leq |z|$ for all $z \in \mathbb{D}$. Moreover, if $|f(z_0)| = |z_0|$ for some $z_0 \in \mathbb{D}$, $z_0 \neq 0$, then there exists $\lambda \in \mathbb{C}$ with $|\lambda| = 1$ such that $f(z) = \lambda z$ for all $z \in \mathbb{D}$.

Proof: Define

$$g(z) = \begin{cases} \frac{f(z)}{z} & \text{if } z \neq 0 \\ f'(0) & \text{if } z = 0. \end{cases}$$

Since $f(0) = 0$, the singularity at 0 is removable, so g is holomorphic on $\mathbb{D}$.

Fix $0 < r < 1$. On the circle $|z| = r$,

$$|g(z)| = \frac{|f(z)|}{|z|} < \frac{1}{r}.$$

By the maximum modulus principle, for $|z| \leq r$ we have $|g(z)| \leq \frac{1}{r}$. Letting $r \rightarrow 1$ gives $|g(z)| \leq 1$ for all $z \in \mathbb{D}$. Hence, for $z \neq 0$,

$$|f(z)| = |z||g(z)| \leq |z|,$$

and the inequality is also true at $z = 0$.

Now suppose $|f(z_0)| = |z_0|$ for some $z_0 \neq 0$. Then $|g(z_0)| = 1$, so $|g|$ has a local maximum in $\mathbb{D}$. By the maximum modulus principle, g is constant. Thus $g(z) = \lambda$ for some $\lambda \in \mathbb{C}$ with $|\lambda| = 1$, and so $f(z) = \lambda z$ for all $z \in \mathbb{D}$. $\square$

6.5 Minimum Modulus Principle

Proposition 6.5

Minimum Modulus Principle

Suppose $G \subseteq \mathbb{C}$ is a region and $f : G \rightarrow \mathbb{C}$ is holomorphic and nonconstant. If $|f|$ has a local minimum at $z_0 \in G$, then $f(z_0) = 0$.

Proof: Suppose $|f|$ has a local minimum at z_0 and $f(z_0) \neq 0$. Then f is nonzero on some disk centered at z_0 , so $\frac{1}{f}$ is holomorphic on that disk. Since $|f|$ has a local minimum at z_0 , $\left|\frac{1}{f}\right|$ has a local maximum at z_0 . By the maximum modulus principle, $\frac{1}{f}$ is constant on a disk centered at z_0 , so f is constant on that disk. By the identity theorem, f is constant on G , a contradiction. Thus $f(z_0) = 0$. $\square$

Example 6.2

The conclusion is possible: if $f(z) = z$ and $K = \overline{\mathbb{D}}$, then $|f|$ attains its minimum at 0, and $f(0) = 0$.

7 Laurent Series

7.1 Examples

Example 7.1

For $|z| < 1$,

$$\frac{1}{1-z} = \sum_{n=0}^{\infty} z^n.$$

For $|z| > 1$,

$$\frac{1}{1-z} = -\frac{1}{z} \cdot \frac{1}{1-\frac{1}{z}} = -\sum_{n=0}^{\infty} z^{-n-1} = -\sum_{n=1}^{\infty} z^{-n}.$$

Example 7.2

Consider

$$\frac{1}{(z-1)(z-2)} = \frac{1}{z-2} - \frac{1}{z-1}.$$

The Laurent expansion depends on the annulus.

If $|z| < 1$, then

$$\begin{aligned} \frac{1}{z-2} &= -\frac{1}{2} \cdot \frac{1}{1-\frac{z}{2}} = -\sum_{n=0}^{\infty} \frac{z^n}{2^{n+1}}, \\ -\frac{1}{z-1} &= \frac{1}{1-z} = \sum_{n=0}^{\infty} z^n. \end{aligned}$$

Hence

$$\frac{1}{(z-1)(z-2)} = \sum_{n=0}^{\infty} \left(1 - \frac{1}{2^{n+1}}\right) z^n.$$

If $1 < |z| < 2$, then

$$\begin{aligned} \frac{1}{z-2} &= -\frac{1}{2} \cdot \frac{1}{1-\frac{z}{2}} = -\sum_{n=0}^{\infty} \frac{z^n}{2^{n+1}}, \\ -\frac{1}{z-1} &= -\frac{1}{z} \cdot \frac{1}{1-\frac{1}{z}} = -\sum_{n=0}^{\infty} z^{-n-1}. \end{aligned}$$

Hence

$$\frac{1}{(z-1)(z-2)} = -\sum_{n=0}^{\infty} \frac{z^n}{2^{n+1}} - \sum_{n=1}^{\infty} z^{-n}.$$

If $|z| > 2$, then

$$\frac{1}{z-2} = \frac{1}{z} \cdot \frac{1}{1-\frac{2}{z}} = \sum_{n=0}^{\infty} 2^n z^{-n-1},$$

$$-\frac{1}{z-1} = -\frac{1}{z} \cdot \frac{1}{1-\frac{1}{z}} = -\sum_{n=0}^{\infty} z^{-n-1}.$$

Hence

$$\frac{1}{(z-1)(z-2)} = \sum_{n=0}^{\infty} (2^n - 1) z^{-n-1}.$$

7.2 Definition and Convergence

Definition 7.1

Laurent Series

A *Laurent Series* centered at z_0 is a series of the form

$$\sum_{n=-\infty}^{\infty} a_n (z - z_0)^n$$

where $a_n \in \mathbb{C}$. This converges at z if both

$$\sum_{n=0}^{\infty} a_n (z - z_0)^n \quad \text{and} \quad \sum_{n=1}^{\infty} a_{-n} (z - z_0)^{-n}$$

converge. Equivalently, the symmetric partial sums

$$\sum_{n=-N}^N a_n (z - z_0)^n$$

converge as $N \rightarrow \infty$. In that case,

$$\lim_{N \rightarrow \infty} \sum_{n=-N}^N a_n (z - z_0)^n$$

is defined to be the sum of the Laurent series.

Theorem 7.1**Convergence of Laurent Series**

Let

$$\sum_{n=-\infty}^{\infty} a_n (z - z_0)^n$$

be a Laurent series. Define

$$R_2 = \frac{1}{\limsup_{n \rightarrow \infty} |a_n|^{\frac{1}{n}}}$$

and

$$R_1 = \limsup_{n \rightarrow \infty} |a_{-n}|^{\frac{1}{n}}.$$

Then the positive part converges absolutely and locally uniformly on

$$|z - z_0| < R_2,$$

and the negative part converges absolutely and locally uniformly on

$$|z - z_0| > R_1.$$

Hence, if $R_1 < R_2$, the Laurent series converges absolutely and locally uniformly on the annulus

$$R_1 < |z - z_0| < R_2,$$

and the symmetric partial sums

$$\sum_{n=-N}^N a_n (z - z_0)^n$$

converge locally uniformly there. This annulus is called the *annulus of convergence* of the Laurent series.

Proof: By the Cauchy-Hadamard theorem, the positive part is an ordinary power series with radius of convergence R_2 , so it converges absolutely and locally uniformly on $|z - z_0| < R_2$.

For the negative part, the root test gives

$$\limsup_{n \rightarrow \infty} |a_{-n} (z - z_0)^{-n}|^{\frac{1}{n}} = \frac{R_1}{|z - z_0|}.$$

Thus the negative part converges absolutely when $\frac{R_1}{|z - z_0|} < 1$, i.e. when $|z - z_0| > R_1$. The same estimate gives locally uniform convergence on compact subsets of $|z - z_0| > R_1$.

On every compact subannulus

$$R_1 < r \leq |z - z_0| \leq R < R_2,$$

both the positive and negative parts converge uniformly. Hence, if $R_1 < R_2$, the Laurent series converges absolutely and the symmetric partial sums converge locally uniformly on $R_1 < |z - z_0| < R_2$. □

Theorem 7.2

Differentiating Laurent Series

Suppose the Laurent series

$$f(z) = \sum_{n=-\infty}^{\infty} a_n(z - z_0)^n$$

converges on the annulus $R_1 < |z - z_0| < R_2$. Then f is holomorphic on this annulus, and

$$f'(z) = \sum_{n=-\infty}^{\infty} n a_n(z - z_0)^{n-1}.$$

Proof: By the convergence theorem, the symmetric partial sums converge locally uniformly on the annulus. Each symmetric partial sum is holomorphic on the annulus, so by Weierstrass's theorem the limit f is holomorphic.

Again by Weierstrass's theorem, the derivatives of the symmetric partial sums converge locally uniformly to f' . Since

$$\frac{d}{dz} \sum_{n=-N}^N a_n(z - z_0)^n = \sum_{n=-N}^N n a_n(z - z_0)^{n-1},$$

it follows that

$$f'(z) = \sum_{n=-\infty}^{\infty} n a_n(z - z_0)^{n-1}.$$

□

7.3 Annuli

Theorem 7.3

Cauchy's Theorem for an Annulus

Suppose $0 \leq R_1 < R_2 \leq \infty$ and f is holomorphic on the annulus

$$\{z \in \mathbb{C} \mid R_1 < |z - z_0| < R_2\}.$$

For $R_1 < r < R_2$, let C_r be the positively oriented circle centered at z_0 with radius r . Then $\int_{C_r} f(z) dz$ is independent of r . In other words, if $R_1 < r < s < R_2$, then

$$\int_{C_r} f(z) dz = \int_{C_s} f(z) dz.$$

Proof: Let

$$I(r) = \int_{C_r} f(z) \, dz.$$

Using the parametrization $z = z_0 + re^{it}$, $0 \leq t \leq 2\pi$, we have

$$I(r) = \int_0^{2\pi} f(z_0 + re^{it}) ire^{it} \, dt.$$

The integrand has continuous partial derivatives in r and t , so we may differentiate under the integral sign:

$$I'(r) = \int_0^{2\pi} \frac{\partial}{\partial r} (f(z_0 + re^{it}) ire^{it}) \, dt.$$

Now

$$\begin{aligned} \frac{\partial}{\partial r} (f(z_0 + re^{it}) ire^{it}) &= f'(z_0 + re^{it}) ire^{2it} + f(z_0 + re^{it}) ie^{it} \\ &= \frac{\partial}{\partial t} (f(z_0 + re^{it}) e^{it}). \end{aligned}$$

Therefore

$$\begin{aligned} I'(r) &= \int_0^{2\pi} \frac{\partial}{\partial t} (f(z_0 + re^{it}) e^{it}) \, dt \\ &= f(z_0 + re^{2\pi i}) e^{2\pi i} - f(z_0 + r) \\ &= f(z_0 + r) - f(z_0 + r) \\ &= 0. \end{aligned}$$

Hence I is constant on (R_1, R_2) , so $\int_{C_r} f(z) \, dz$ is independent of r . □

7.4 Cauchy's Formula on an Annulus

Theorem 7.4

Cauchy's Formula for an Annulus

Suppose $0 \leq R_1 < R_2 \leq \infty$ and f is holomorphic on the annulus

$$\{z \in \mathbb{C} \mid R_1 < |z - z_0| < R_2\}.$$

For $R_1 < r < R_2$, let C_r be the positively oriented circle centered at z_0 with radius r . If

$$R_1 < r_1 < |w - z_0| < r_2 < R_2,$$

then

$$f(w) = \frac{1}{2\pi i} \int_{C_{r_2}} \frac{f(z)}{z - w} \, dz - \frac{1}{2\pi i} \int_{C_{r_1}} \frac{f(z)}{z - w} \, dz.$$

Proof: Fix w as above and define

$$g(z) = \begin{cases} \frac{f(z)-f(w)}{z-w} & \text{if } z \neq w \\ f'(w) & \text{if } z = w. \end{cases}$$

Then g is holomorphic on the annulus. By Cauchy's theorem for an annulus,

$$\int_{C_{r_1}} g(z) \, dz = \int_{C_{r_2}} g(z) \, dz.$$

Hence

$$\int_{C_{r_2}} \frac{f(z)}{z-w} \, dz - \int_{C_{r_1}} \frac{f(z)}{z-w} \, dz = f(w) \int_{C_{r_2}} \frac{1}{z-w} \, dz - f(w) \int_{C_{r_1}} \frac{1}{z-w} \, dz.$$

The first integral on the right is $2\pi i$ by Cauchy's formula for the circle, since w lies inside C_{r_2} . The second integral is 0 by Cauchy's theorem for convex regions, since w lies outside C_{r_1} and $z \mapsto \frac{1}{z-w}$ is holomorphic on the disk bounded by C_{r_1} . Therefore

$$\int_{C_{r_2}} \frac{f(z)}{z-w} \, dz - \int_{C_{r_1}} \frac{f(z)}{z-w} \, dz = 2\pi i f(w),$$

which gives the formula. □